My Completed Void CTF Challenges

Name: Mohd Saif
Event: Void CTF
Total Challenges: 8

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1. 3000 Files Later

Category: Forensics
Points: 500

Flag: VOID{y0u_ch3ck3d_3000_f1l3s_f0r_th1s_021959f6}

Approach:

- Successfully retrieved the challenge flag.
- Detailed file-analysis steps have not yet been documented.

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2. Robots That Lie

Category: Web Exploitation
Points: 100

Flag: VOID{r0b0ts_c4nt_k33p_s3cr3ts}

Approach:

1. Opened the website's robots.txt file.
2. Examined the listed paths, particularly the Disallow entries.
3. Identified the hidden destination from the paths.
4. Opened the destination in the browser.
5. Retrieved the flag from the access-granted page.

Key Learning: robots.txt may reveal paths that are not intended for crawlers, but it does not provide access control.

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3. Fractured Truth

Category: Steganography
Points: 300

Flag: VOID{s33_th3_wh0l3_p1ctur3}

Approach:

1. Extracted the archive and found 12 PNG fragments.
2. Identified that the fragments belonged to two different QR codes.
3. Compared tile edges to determine possible arrangements.
4. Reconstructed the QR codes and decoded both messages.
5. Selected the flag that matched the challenge's reconstruction theme rather than the apparent decoy.

Key Learning: Image reconstruction and QR decoding can recover information hidden across fragmented images.

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4. Transmission

Category: Miscellaneous / Audio Steganography
Points: 300

Flag: VOID{n0_v1s_h4lf_th3_st0ry}

Approach:

1. Inspected the WAV metadata and found a hexadecimal note.
2. Examined the audio spectrogram and identified an SSTV signal.
3. Decoded the audio in Martin M1 mode to recover the first part of the flag.
4. Used the hint MIRA to XOR-decode the hexadecimal note.
5. Joined the recovered parts to obtain the complete flag.

Key Learning: Audio can encode images and messages, while metadata can contain additional clues.

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5. Volatile Mira

Category: Forensics / Memory Forensics
Points: 300

Flag: VOID{f0r3n51c5_15_4n_4r7}

Approach:

1. Extracted the archive and obtained a LiME memory image.
2. Searched memory strings and identified relevant Python processes and a session token.
3. Used a shell-history clue to search for fromhex in memory.
4. Located a script containing a hexadecimal string named SEALED.
5. Identified the repeating XOR key from the session token.
6. Decrypted the string to recover the flag.

Key Learning: Memory images can preserve process information and script contents useful for digital investigations.

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6. Bing Chilling

Category: Reverse Engineering

Flag: VOID{1_l1k3_ic3_cr34m_t00!!!!}

Approach:

1. Identified the challenge binary as a stripped LoongArch64 ELF.
2. Reconstructed the program's character-transformation logic.
3. Used the supplied numerical output to reverse the transformations.
4. Tested candidate byte values for each position.
5. Combined the recovered characters to obtain the flag.

Key Learning: Understanding a binary's transformation logic can help reverse program-generated output.

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7. Now You See Mee

Category: Web Exploitation
Points: 300

Flag: VOID{g0ph3r_r3sp_sl1pp3d_past_a_sl33py_waf}

Approach:

- The flag was confirmed from the shared challenge screenshot.
- The exact exploitation steps have not yet been documented.

Key Learning: Web exploitation challenges can test how applications process requests and how security controls handle unusual inputs.

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8. Archive_042

Category: Cryptography
Points: 200

Flag: Not provided in the available notes.

Approach:

- The challenge is included in the completed-challenges list.
- The exact flag and solving steps have not yet been documented.

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